

# Homework 4

## Problem 1

(a) Note the following expansion, which is used frequently in Astrodynamics.

$$\frac{1}{|\mathbf{r} - \mathbf{r}_p|} = \frac{1}{r_p} \sum_{i=0}^{\infty} \left(\frac{r}{r_p}\right)^i P_{i,0}(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}_p),$$

where  $P_{i,0}(x)$  are known as Legendre polynomials, expressed using the Rodrigues formula as:

$$P_{i,0}(x) = \frac{1}{2^i i!} \frac{d^i}{dx^i} (x^2 - 1)^i.$$

Show that the perturbation potential equals, to the lowest non-trivial order,

$$\mathcal{R}_p \approx \frac{\mu_p}{2r_p^3} [3(\mathbf{r} \cdot \hat{\mathbf{r}}_p)^2 - r^2],$$

where a term independent of the dynamical variable  $\mathbf{r}$  can be dropped (*explain why*). This is referred to as the quadrupolar- or Hill-approximated third-body disturbing function.

$$\Rightarrow \frac{1}{|\mathbf{r} - \mathbf{r}_p|} \approx \frac{1}{r_p} \left[ 1 + \frac{r}{r_p} \cos \psi + \left(\frac{r}{r_p}\right)^2 \frac{1}{2} (3 \cos^2 \psi - 1) \right] = \frac{1}{r_p} + \frac{r \cos \psi}{r_p^2} + \frac{r^2 (3 \cos^2 \psi - 1)}{2r_p^3}$$

$$\begin{aligned} \cos \psi &= \hat{\mathbf{r}} \cdot \hat{\mathbf{r}}_p \Rightarrow r \cos \psi = \mathbf{r} \cdot \hat{\mathbf{r}}_p \\ P_0(\cos \psi) &= 1 \\ P_1(\cos \psi) &= \cos \psi \\ P_2(\cos \psi) &= \frac{1}{2} (3 \cos^2 \psi - 1) \end{aligned}$$

$$\Rightarrow \mathcal{R}_p = \mu_p \left[ \frac{1}{|\mathbf{r} - \mathbf{r}_p|} - \frac{\mathbf{r}_p \cdot \mathbf{r}}{r_p^3} \right] \approx \mu_p \left[ \frac{1}{r_p} + \frac{r \cos \psi}{r_p^2} + \frac{r^2 (3 \cos^2 \psi - 1)}{2r_p^3} - \frac{\mathbf{r}_p \cdot \mathbf{r}}{r_p^3} \right] \approx \mu_p \left[ \frac{1}{r_p} + \frac{r \cos \psi}{r_p^2} + \frac{r^2 (3 \cos^2 \psi - 1)}{2r_p^3} - \frac{\hat{\mathbf{r}}_p \cdot \mathbf{r}}{r_p^2} \right]$$

$$\approx \mu_p \left[ \frac{1}{r_p} + \frac{r^2 (3 \cos^2 \psi - 1)}{2r_p^3} \right]$$

$$\approx \mu_p \left[ \frac{3(\mathbf{r} \cdot \hat{\mathbf{r}}_p)^2 - r^2}{2r_p^3} \right]$$

$(r \cos \psi)^2 = (\mathbf{r} \cdot \hat{\mathbf{r}}_p)^2$   $\mu_p/r_p$  term can be dropped because it does not depend on the spacecraft position  $\mathbf{r}$ . Therefore its gradient w/ respect to  $\mathbf{r}$  is zero, so it produces no disturbing acceleration

$$\mathcal{R}_p \approx \frac{\mu_p}{2r_p^3} [3(\mathbf{r} \cdot \hat{\mathbf{r}}_p)^2 - r^2]$$

(b) We can also express the first term in the general third-body disturbing function as

$$\frac{1}{|\mathbf{r} - \mathbf{r}_p|} = \frac{1}{r_p} \left[ 1 + \left(\frac{r}{r_p}\right)^2 - 2 \left(\frac{r}{r_p}\right) \hat{\mathbf{r}} \cdot \hat{\mathbf{r}}_p \right]^{-1/2}$$

Recall the binomial theorem,

$$\begin{aligned} (x + y)^r &= \sum_{k=0}^{\infty} \binom{r}{k} x^{r-k} y^k \\ &= x^r + r x^{r-1} y + \frac{r(r-1)}{2!} x^{r-2} y^2 + \dots \end{aligned}$$

Neglecting terms of  $\mathcal{O}(r/r_p)^3$  and higher (i.e., assuming  $r/r_p \ll 1$ ), expand the first term in  $\mathcal{R}_p$  using the binomial theorem and show that we recover the Hill, or quadrupole, approximation.

$$\alpha = \frac{r}{r_p}, \quad c = \hat{\mathbf{r}} \cdot \hat{\mathbf{r}}_p$$

$$\frac{1}{|\mathbf{r} - \mathbf{r}_p|} = \frac{1}{r_p} \left[ 1 + \alpha^2 - 2\alpha c \right]^{-1/2}$$

$$(1+x)^{-1/2} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \text{H.O.T.}$$

$$(1+x)^{-1/2} \approx 1 - \frac{1}{2}(\alpha^2 - 2\alpha c) + \frac{3}{8}(\alpha^2 - 2\alpha c)^2$$

$$\Rightarrow (1+x)^{-1/2} \approx 1 - \frac{1}{2}\alpha^2 + \alpha c + \frac{3}{8}(2\alpha^4 - 4\alpha^3 c + 4\alpha^2 c^2) \approx 1 - \frac{1}{2}\alpha^2 + \alpha c + \frac{3}{2}\alpha^2 c^2 \approx 1 + \alpha c + \frac{1}{2}\alpha^2(3c^2 - 1)$$

$$\Rightarrow \frac{1}{|\mathbf{r} - \mathbf{r}_p|} \approx \frac{1}{r_p} \left[ 1 + \alpha c + \frac{1}{2}\alpha^2(3c^2 - 1) \right] = \frac{1}{r_p} \left[ 1 + \frac{r}{r_p} c + \frac{1}{2} \left(\frac{r}{r_p}\right)^2 (3c^2 - 1) \right]$$

$$\Rightarrow \mathcal{R}_p = \mu_p \left[ \frac{1}{|\mathbf{r} - \mathbf{r}_p|} - \frac{\mathbf{r}_p \cdot \mathbf{r}}{r_p^3} \right] \approx \mu_p \left[ \frac{1}{r_p} + \frac{r c}{r_p^2} + \frac{r^2 (3c^2 - 1)}{2r_p^3} - \frac{\mathbf{r}_p \cdot \mathbf{r}}{r_p^3} \right] \approx \mu_p \left[ \frac{1}{r_p} + \frac{r \hat{\mathbf{r}}_p \cdot \mathbf{r}}{r_p^3} + \frac{r^2 (3c^2 - 1)}{2r_p^3} - \frac{\hat{\mathbf{r}}_p \cdot \mathbf{r}}{r_p^2} \right]$$

$$\approx \mu_p \left[ \frac{r^2 (3(\hat{\mathbf{r}}_p \cdot \hat{\mathbf{r}})^2 - 1)}{2r_p^3} \right] \approx \frac{\mu_p}{2r_p^3} [3(\mathbf{r} \cdot \hat{\mathbf{r}}_p)^2 - r^2]$$

ignore because the gradient will equal zero

(c) Write out the disturbing acceleration that corresponds to this truncated potential.

$$\begin{aligned} \ddot{\mathbf{a}}_d &= \frac{d\mathcal{R}_p}{dr} = \frac{d}{dr} \left( \frac{3\mu_p}{2r_p^3} (\mathbf{r} \cdot \hat{\mathbf{r}}_p)^2 - \frac{\mu_p r^2}{2r_p^3} \right) \\ &= \frac{3\mu_p}{2r_p^3} 2(\mathbf{r} \cdot \hat{\mathbf{r}}_p) \hat{\mathbf{r}}_p - \frac{\mu_p r}{r_p^3} \end{aligned}$$

$$\ddot{\mathbf{a}}_d = \frac{\mu_p}{r_p^3} (3(\mathbf{r} \cdot \hat{\mathbf{r}}_p) \hat{\mathbf{r}}_p - \mathbf{r}) \rightarrow \text{Hill/quadrupole third-body disturbing acceleration}$$

## Problem 2

2. Consider the perturbed Kepler problem for a constant disturbing acceleration  $\mathbf{a}_s$ :

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_s.$$

- (a) Form the perturbation function  $\mathcal{R}$  in terms of the osculating orbital elements, for a constant acceleration acting along the  $z$ -axis. It should have the form

$$\mathcal{R} = r \sin i \sin(\omega + f) a_s,$$

where  $r = a(1 - e^2)/(1 + e \cos f)$  is the radius function (trajectory equation),  $a$  is the semi-major axis,  $e$  is the eccentricity,  $i$  is the inclination,  $\omega$  is the argument of periapsis,  $f$  is the true anomaly, and  $a_s$  is the acceleration magnitude.

$$\Rightarrow \vec{a}_s = a_s \hat{z} \quad \vec{a}_s = \begin{bmatrix} \frac{\partial \mathcal{R}}{\partial x} \\ \frac{\partial \mathcal{R}}{\partial y} \\ \frac{\partial \mathcal{R}}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ a_s \end{bmatrix} \Rightarrow R = a_s z \quad \vec{r} = \begin{bmatrix} r \cos f \\ r \sin f \\ 0 \end{bmatrix} \quad \vec{r}(f) = r [\cos f \hat{x} + \sin f (\hat{h} \times \hat{e})]$$

transformation from orbital plane to the inertial frame  $\vec{r} = R(L)R(i)R(\omega) \begin{bmatrix} r \cos f \\ r \sin f \\ 0 \end{bmatrix}$    
  $\nearrow$  rotation about  $z$ :  $z$ -component remains the same

$$R(\omega) \begin{bmatrix} r \cos f \\ r \sin f \\ 0 \end{bmatrix} = \begin{bmatrix} r \cos(\omega + f) \\ r \sin(\omega + f) \\ 0 \end{bmatrix}, \quad R(i) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & -\sin i \\ 0 & \sin i & \cos i \end{bmatrix} \Rightarrow R(i)R(\omega) \begin{bmatrix} r \cos f \\ r \sin f \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & -\sin i \\ 0 & \sin i & \cos i \end{bmatrix} \begin{bmatrix} r \cos(\omega + f) \\ r \sin(\omega + f) \\ 0 \end{bmatrix} = \begin{bmatrix} r \cos(\omega + f) \\ r \cos(\omega + f) \cos i \\ r \sin(\omega + f) \sin i \end{bmatrix}$$

$\therefore$  the inertial  $z$ -component is  $z = r \sin i \sin(\omega + f)$

$$\Rightarrow R = a_s z = a_s r \sin i \sin(\omega + f) = a_s \left( \frac{a(1-e^2) \sin i \sin(\omega + f)}{1 + e \cos f} \right)$$

- (b) Write out the Lagrange equations for the osculating semi-major axis  $a$  and eccentricity  $e$  for the unaveraged case. You should find the following:

From lecture slides:

$$\dot{a} = \frac{2}{na} \frac{d\mathcal{R}}{d\sigma}$$

$$\dot{e} = \frac{\sqrt{1-e^2}}{na^2 e} \left[ \sqrt{1-e^2} \frac{d\mathcal{R}}{d\sigma} - \frac{d\mathcal{R}}{d\omega} \right]$$

Note that to compute  $\partial \mathcal{R} / \partial \sigma$  will require use of the chain rule to find

$$\frac{\partial \mathcal{R}}{\partial \sigma} = \frac{\partial \mathcal{R}}{\partial f} \frac{\partial f}{\partial E} \frac{\partial E}{\partial \sigma}.$$

Trig Identities  $\sin(A+B) = \sin A \cos B + \cos A \sin B$

$$\Rightarrow r \sin(\omega + f) = r \sin \omega \cos f + r \cos \omega \sin f \quad r \cos f = a(\cos E - e), \quad r \sin f = a\sqrt{1-e^2} \sin E$$

$$= a \sin \omega (\cos E - e) + a\sqrt{1-e^2} \cos \omega \sin E$$

$$\Rightarrow R = a_s \sin i (a \sin \omega (\cos E - e) + a\sqrt{1-e^2} \cos \omega \sin E) \Rightarrow R = a_s a \sin i (\sin \omega (\cos E - e) + \sqrt{1-e^2} \cos \omega \sin E) \quad \sigma - M$$

$$\Rightarrow \frac{d\mathcal{R}}{d\sigma} = \frac{d\mathcal{R}}{dM} \quad M = E - e \sin E \quad \frac{dM}{dE} = 1 - e \cos E \Rightarrow \frac{dE}{dM} = \frac{1}{1 - e \cos E}$$

$$\Rightarrow \frac{d\mathcal{R}}{dE} = a_s a \sin i (-\sin \omega \sin E + \sqrt{1-e^2} \cos \omega \cos E) \Rightarrow \frac{d\mathcal{R}}{dM} = \frac{d\mathcal{R}}{dE} \frac{dE}{dM}$$

$$\Rightarrow \frac{d\mathcal{R}}{d\sigma} = a_s a \sin i (-\sin \omega \sin E + \sqrt{1-e^2} \cos \omega \cos E) \frac{1}{1 - e \cos E} \quad \sin E = \frac{\sqrt{1-e^2} \sin f}{1 + e \cos f}, \quad \cos E = \frac{e + \cos f}{1 + e \cos f}, \quad 1 - e \cos E = \frac{1 - e^2}{1 + e \cos f}$$

$$= a_s a \sin i \left( -\sin \omega \frac{\sqrt{1-e^2} \sin f}{1 + e \cos f} + \sqrt{1-e^2} \cos \omega \frac{e + \cos f}{1 + e \cos f} \right) \frac{1 + e \cos f}{1 - e^2}$$

$$= \frac{a_s a \sin i}{\sqrt{1-e^2}} (-\sin \omega \sin f + \cos \omega (e + \cos f)) \quad \cos \omega \cos f - \sin \omega \sin f = \cos(\omega + f)$$

$$\frac{d\mathcal{R}}{d\sigma} = \frac{a_s a \sin i}{\sqrt{1-e^2}} (\cos(\omega + f) + e \cos \omega)$$

$$\Rightarrow \frac{d\mathcal{R}}{d\omega} = a_s r \sin i \cos(\omega + f) = \frac{a_s a (1-e^2) \sin i \cos(\omega + f)}{1 + e \cos f}$$

$$\dot{a} = \frac{2}{na} \frac{d\mathcal{R}}{d\sigma} = \frac{2a_s \sin i}{n\sqrt{1-e^2}} (\cos(\omega + f) + e \cos \omega)$$

$$\dot{e} = \frac{\sqrt{1-e^2}}{na^2 e} \left[ \sqrt{1-e^2} \frac{d\mathcal{R}}{d\sigma} - \frac{d\mathcal{R}}{d\omega} \right], \quad \beta = \sqrt{1-e^2}$$

$$\dot{e} = \frac{\beta}{na^2 e} \left( \beta \frac{dR}{d\sigma} - \frac{dR}{dw} \right) \quad \beta \frac{dR}{d\sigma} = a_s a \sin i (\cos(w+f) + e \cos w), \quad \frac{dR}{dw} = \frac{a_s a \beta^2 \sin i \cos(w+f)}{1 + e \cos f}$$

$$\begin{aligned} \beta \frac{dR}{d\sigma} - \frac{dR}{dw} &= a_s a \sin i (\cos(w+f) + e \cos w) - \frac{a_s a \beta^2 \sin i \cos(w+f)}{1 + e \cos f} \\ &= a_s a \sin i \left( \cos(w+f) + e \cos w - \frac{\beta^2 \cos(w+f)}{1 + e \cos f} \right) \\ &= a_s a \sin i \left( e \cos w + \left( 1 - \frac{\beta^2}{1 + e \cos f} \right) \cos(w+f) \right) \quad \beta^2 = 1 - e^2 \quad 1 - \frac{1 - e^2}{1 + e \cos f} = \frac{1 + e \cos f - 1 + e^2}{1 + e \cos f} \\ &= a_s a \sin i \left( e \cos w + \left( \frac{e \cos f + e^2}{1 + e \cos f} \right) \cos(w+f) \right) \end{aligned}$$

$$\dot{e} = \frac{\beta}{na^2 e} \left( a_s a \sin i \left( e \cos w + \left( \frac{e \cos f + e^2}{1 + e \cos f} \right) \cos(w+f) \right) \right)$$

$$= \frac{a_s \sqrt{1 - e^2} \sin i}{na} \left( \cos w + \frac{e + \cos f}{1 + e \cos f} \cos(w+f) \right)$$

### Problem 3

3. Implement a numerical integration scheme (using the language or program of your choice) that:

- Integrate  $\ddot{\mathbf{r}} = -\mu/r^3 \mathbf{r} + \mathbf{a}_d$  for a given set of initial conditions,  $\mathbf{r}_0$  and  $\mathbf{v}_0$  specified at time  $t_0$ , and for a specified constant vector  $\mathbf{a}_d$ .
- Given a numerically integrated trajectory, plot the resultant position and velocity as a function of time, compute and plot all of the classical orbital elements ( $a, e, i, \Omega, \omega, M$ ) as a function of time along the trajectory. Note that you can check your program by setting  $\mathbf{a}_d = \mathbf{0}$  and verifying that all of the orbital elements (except  $M$ ) are constant.

Check the code below

Using this program, implement the following parameters and initial conditions, plot the resulting trajectories and orbital elements, and comment on your results:

$$\begin{aligned} \mu &= 1, \\ \mathbf{r}_0 &= [1, 0, 0], \\ \mathbf{v}_0 &= [0, 1, 0], \\ \mathbf{a}_d &= [a, 0, 0], \end{aligned}$$

where we let  $a$  equal the following values in turn:  $a = 0.001, 0.01, 0.1$ , and  $1$ . For each case plot the trajectory for at least 10 revolutions.

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In [2]: import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# MAE 240 - Homework 4 - Problem 3
mu = 1.0

# Initial Conditions
r0 = np.array([1.0, 0.0, 0.0])
v0 = np.array([0.0, 1.0, 0.0])

# Disturbing acceleration magnitudes
a_values = [0.001, 0.01, 0.1, 1.0]

# Circular orbit period
T = 2 * np.pi

# Simulate for 10 revolutions
t_final = 10 * T

# Time vector
t_eval = np.linspace(0, t_final, 3000)

# Equations of Motion
def eom(t, state, mu, a_d):

    r = state[0:3]
    v = state[3:6]

    r_norm = np.linalg.norm(r)

    # Central gravity acceleration
    a_gravity = -mu * r / r_norm**3

    # Total acceleration
    a_total = a_gravity + a_d

    return np.hstack((v, a_total))

# Cartesian State -> Orbital Elements
def cartesian_to_elements(r, v, mu):

    r_norm = np.linalg.norm(r)
    v_norm = np.linalg.norm(v)

    # Angular Momentum
    h_vec = np.cross(r, v)
    h = np.linalg.norm(h_vec)

    # Inclination
    i = np.arccos(h_vec[2] / h)

    # Node Vector

```



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k_hat = np.array([0.0, 0.0, 1.0])

n_vec = np.cross(k_hat, h_vec)
n = np.linalg.norm(n_vec)

# Eccentricity Vector
e_vec = np.cross(v, h_vec) / mu - r / r_norm
e = np.linalg.norm(e_vec)

# Specific Orbital Energy
energy = v_norm**2 / 2 - mu / r_norm

# Semi-major Axis
if abs(energy) > 1e-12:
    a = -mu / (2 * energy)
else:
    a = np.nan

# RAAN
if n > 1e-12:

    Omega = np.arccos(n_vec[0] / n)

    if n_vec[1] < 0:
        Omega = 2 * np.pi - Omega

else:
    Omega = 0.0

# Argument of Periapsis
if n > 1e-12 and e > 1e-12:

    omega = np.arccos(
        np.dot(n_vec, e_vec) / (n * e)
    )

    if e_vec[2] < 0:
        omega = 2 * np.pi - omega

else:
    omega = 0.0

# True Anomaly
if e > 1e-12:

    f = np.arccos(
        np.dot(e_vec, r) / (e * r_norm)
    )

    if np.dot(r, v) < 0:
        f = 2 * np.pi - f

else:
    f = 0.0

# Mean Anomaly

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```

if e < 1 and e > 1e-12:

    E = 2 * np.arctan2(
        np.sqrt(1 - e) * np.sin(f / 2),
        np.sqrt(1 + e) * np.cos(f / 2)
    )

    M = E - e * np.sin(E)

    M = M % (2 * np.pi)

else:
    M = 0.0

return a, e, i, Omega, omega, M

# Run Simulation Cases
for accel_mag in a_values:

    print(f"Running case: a_d = [{accel_mag}, 0, 0]")

    # Disturbing acceleration
    a_d = np.array([accel_mag, 0.0, 0.0])

    # Initial state vector
    state0 = np.hstack((r0, v0))

    # Numerical integration
    sol = solve_ivp(
        eom,
        [0, t_final],
        state0,
        t_eval=t_eval,
        args=(mu, a_d),
        rtol=1e-10,
        atol=1e-12
    )

    t = sol.t
    X = sol.y.T

    r_hist = X[:, 0:3]
    v_hist = X[:, 3:6]

    # Compute Orbital Elements
    elements = np.zeros((len(t), 6))

    for k in range(len(t)):

        elements[k, :] = cartesian_to_elements(
            r_hist[k, :],
            v_hist[k, :],
            mu
        )

    a_hist = elements[:, 0]

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e_hist = elements[:, 1]
i_hist = np.rad2deg(elements[:, 2])
Omega_hist = np.rad2deg(elements[:, 3])
omega_hist = np.rad2deg(elements[:, 4])
M_hist = np.rad2deg(elements[:, 5])

r_mag = np.linalg.norm(r_hist, axis=1)
v_mag = np.linalg.norm(v_hist, axis=1)

# Figure 1: Trajectory
plt.figure(figsize=(7, 7))

plt.plot(
    r_hist[:, 0],
    r_hist[:, 1],
    linewidth=1.5,
    label='Trajectory'
)

plt.scatter(
    r_hist[0, 0],
    r_hist[0, 1],
    marker='o',
    s=50,
    label='Start'
)

plt.scatter(
    0,
    0,
    marker='x',
    s=120,
    label='Central Body'
)

plt.xlabel('x', fontsize=12)
plt.ylabel('y', fontsize=12)

plt.title(
    f'Trajectory for $a_d = [{accel_mag}, 0, 0]$',
    fontsize=16
)

plt.axis('equal')

plt.grid(True)
plt.legend(fontsize=10)

plt.tight_layout()

plt.show()

# Figure 2: Orbital Elements vs Time
fig, axs = plt.subplots(
    4,
    2,

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        figsize=(16, 14),
        sharex=True
    )

    # Main figure title
    fig.suptitle(
        f'Orbital Elements vs Time for '
        f'$a_d = [{accel_mag}, 0, 0]$',
        fontsize=18,
        y=0.98
    )

    # LEFT COLUMN
    # Semi-major Axis

    axs[0, 0].plot(
        t,
        a_hist,
        linewidth=1.5
    )

    axs[0, 0].set_title('Semi-major Axis')
    axs[0, 0].set_ylabel('a')

    # Eccentricity
    axs[1, 0].plot(
        t,
        e_hist,
        linewidth=1.5
    )

    axs[1, 0].set_title('Eccentricity')
    axs[1, 0].set_ylabel('e')

    # Inclination
    axs[2, 0].plot(
        t,
        i_hist,
        linewidth=1.5
    )

    axs[2, 0].set_title('Inclination')
    axs[2, 0].set_ylabel('i [deg]')

    # RAAN
    axs[3, 0].plot(
        t,
        Omega_hist,
        linewidth=1.5
    )

    axs[3, 0].set_title('RAAN')
    axs[3, 0].set_ylabel('$\Omega$ [deg]')
    axs[3, 0].set_xlabel('Time')

    # RIGHT COLUMN

```

```

# Argument of Periapsis
axs[0, 1].plot(
    t,
    omega_hist,
    linewidth=1.5
)

axs[0, 1].set_title('Argument of Periapsis')
axs[0, 1].set_ylabel('ω [deg]')

# Mean Anomaly
axs[1, 1].plot(
    t,
    M_hist,
    linewidth=1.5
)

axs[1, 1].set_title('Mean Anomaly')
axs[1, 1].set_ylabel('M [deg]')

# Radius Magnitude
axs[2, 1].plot(
    t,
    r_mag,
    linewidth=1.5
)

axs[2, 1].set_title('Radius Magnitude')
axs[2, 1].set_ylabel('|r|')

# Velocity Magnitude
axs[3, 1].plot(
    t,
    v_mag,
    linewidth=1.5
)

axs[3, 1].set_title('Velocity Magnitude')
axs[3, 1].set_ylabel('|v|')
axs[3, 1].set_xlabel('Time')

# Formatting
for row in axs:

    for ax in row:

        ax.grid(True)

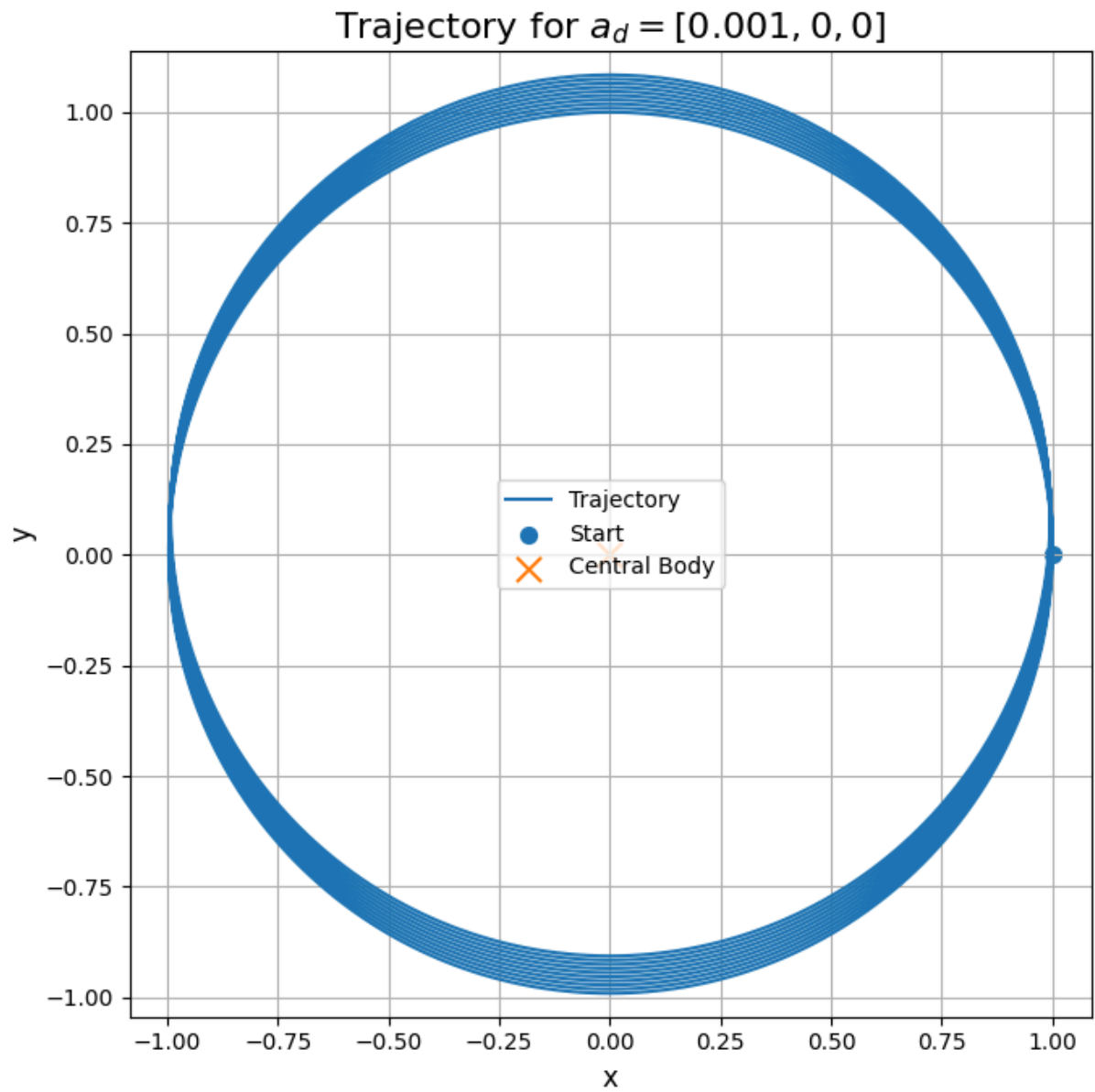
        ax.tick_params(
            axis='both',
            labelsiz=10
        )

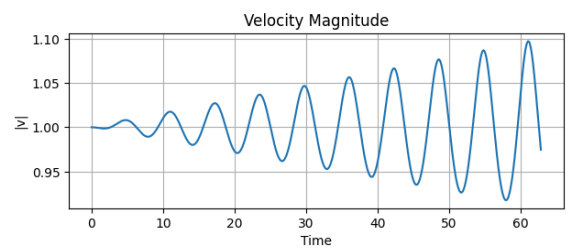
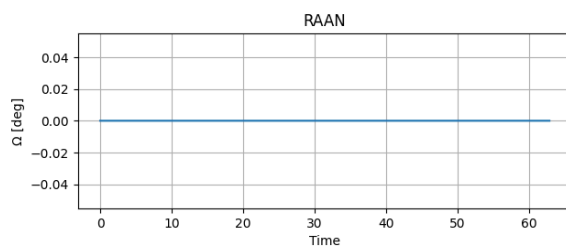
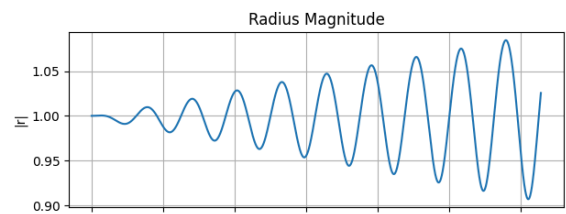
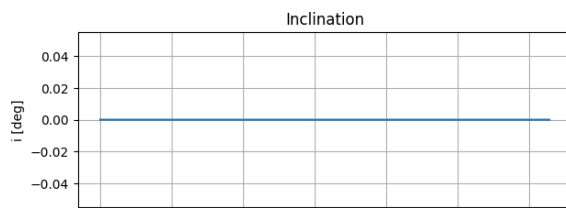
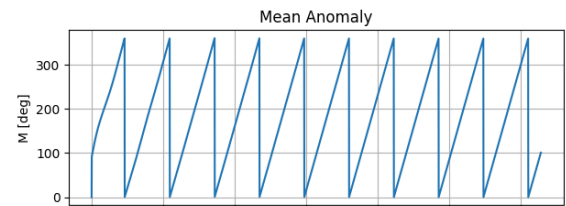
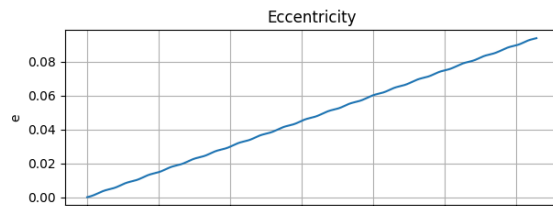
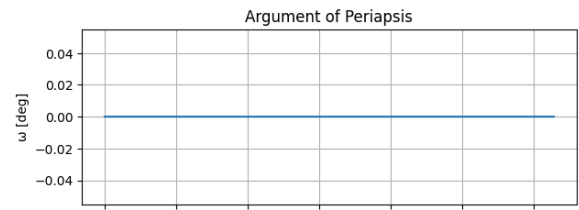
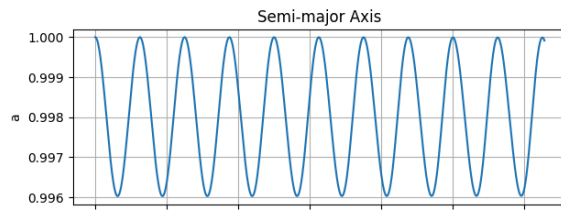
# Adjust subplot spacing
fig.subplots_adjust(
    hspace=0.55,

```

```
wspace=0.30,  
top=0.88  
)  
  
plt.show()
```

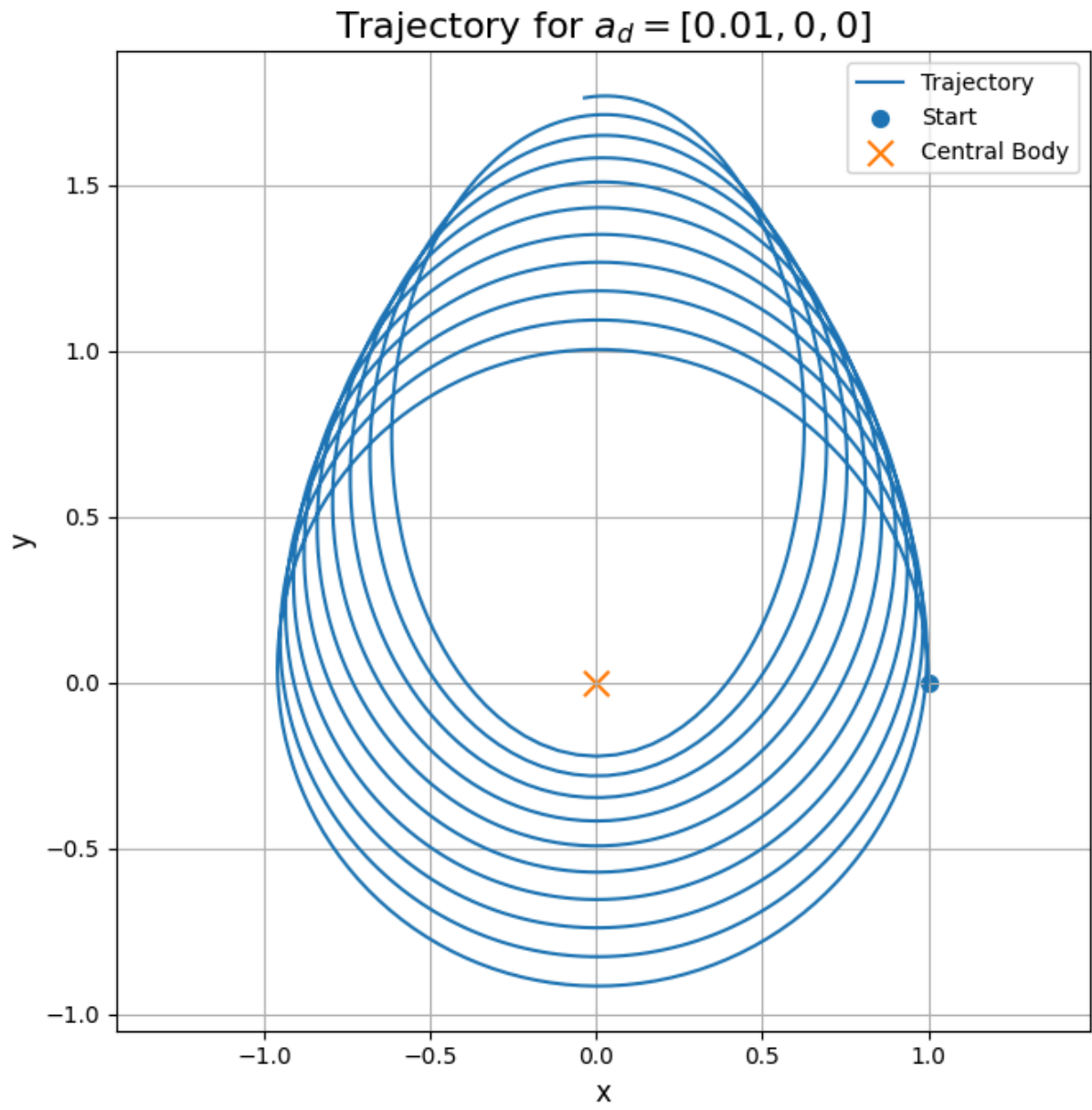
Running case:  $a_d = [0.001, 0, 0]$

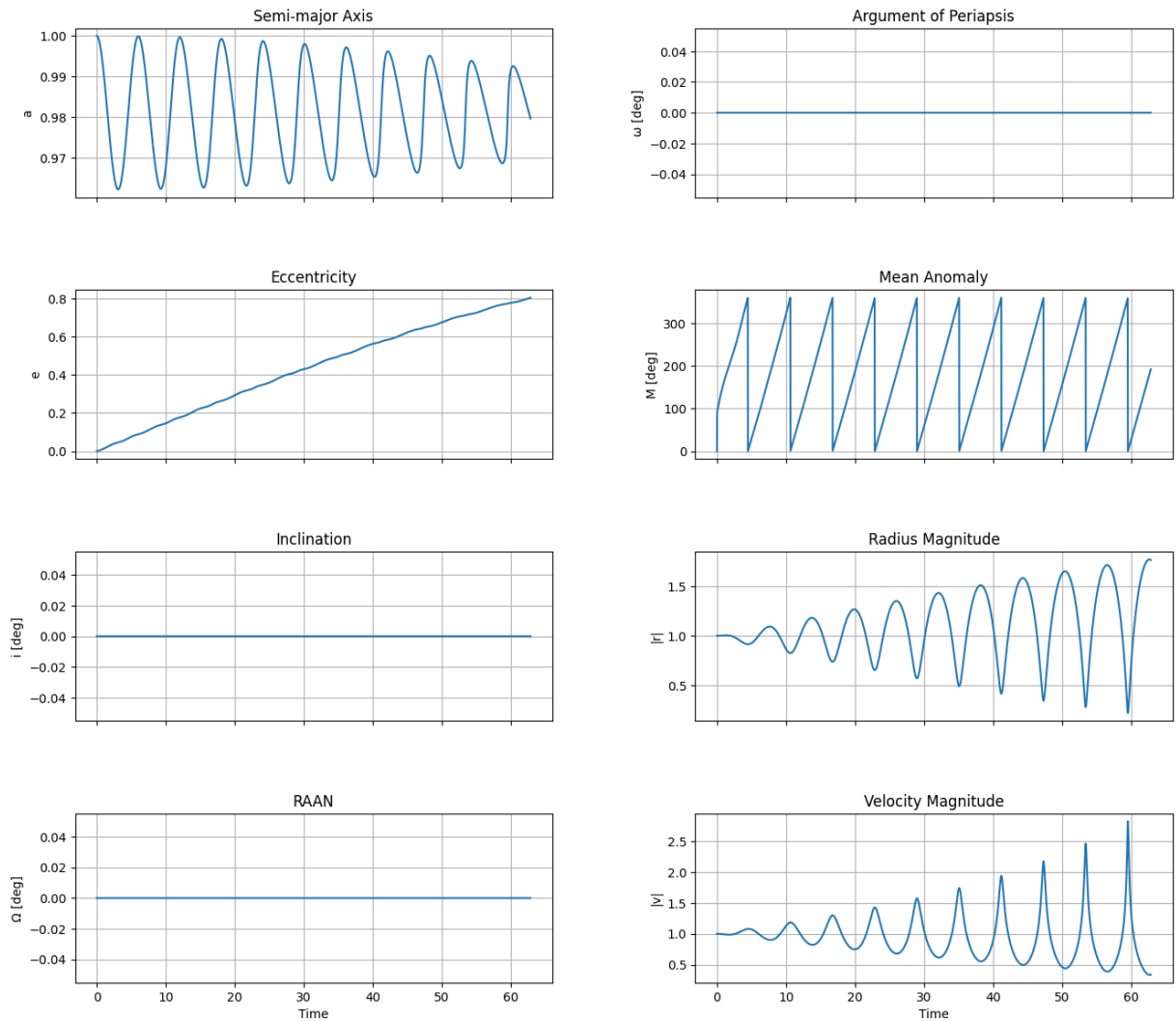


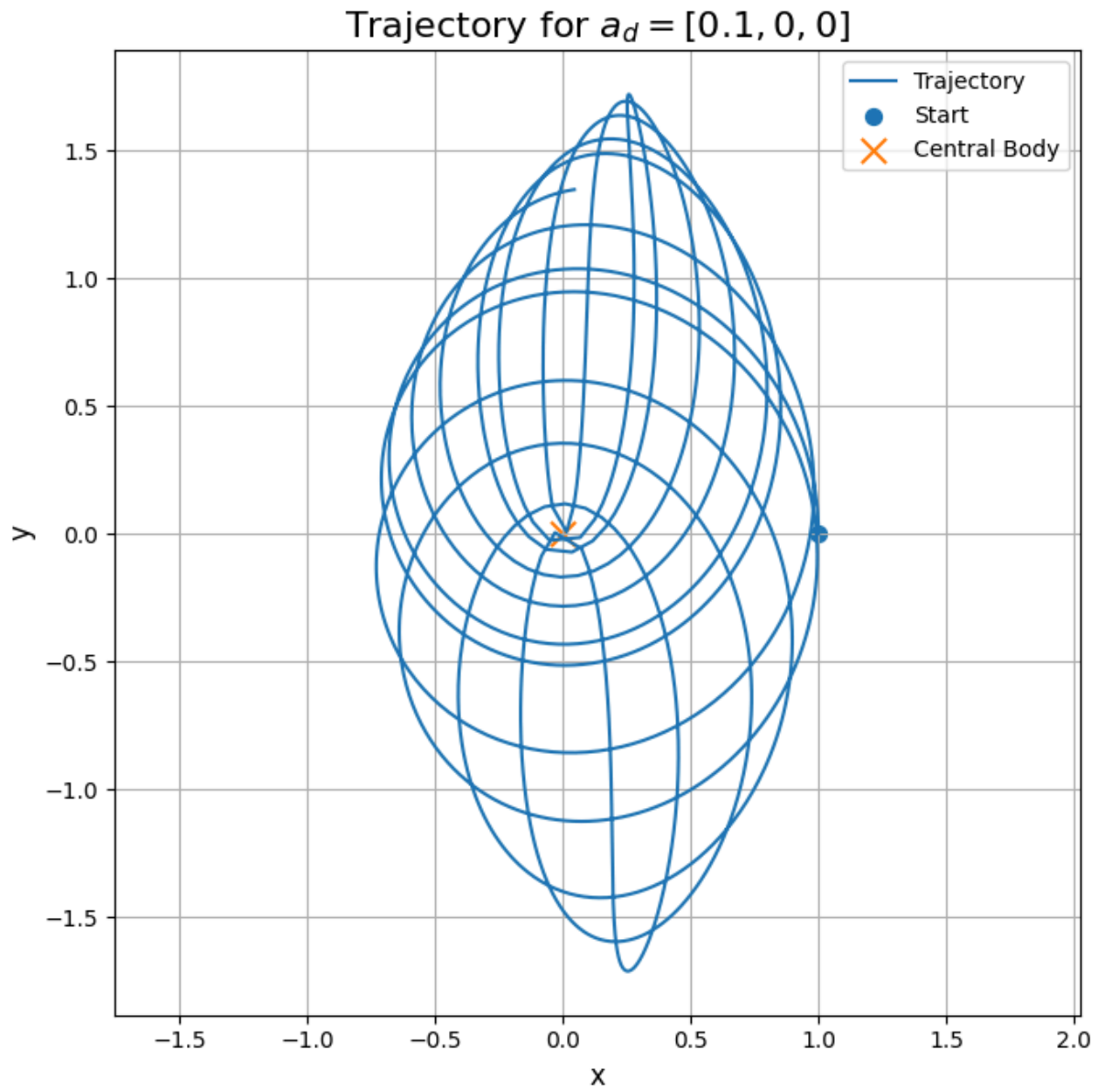
Orbital Elements vs Time for  $a_d = [0.001, 0, 0]$ 

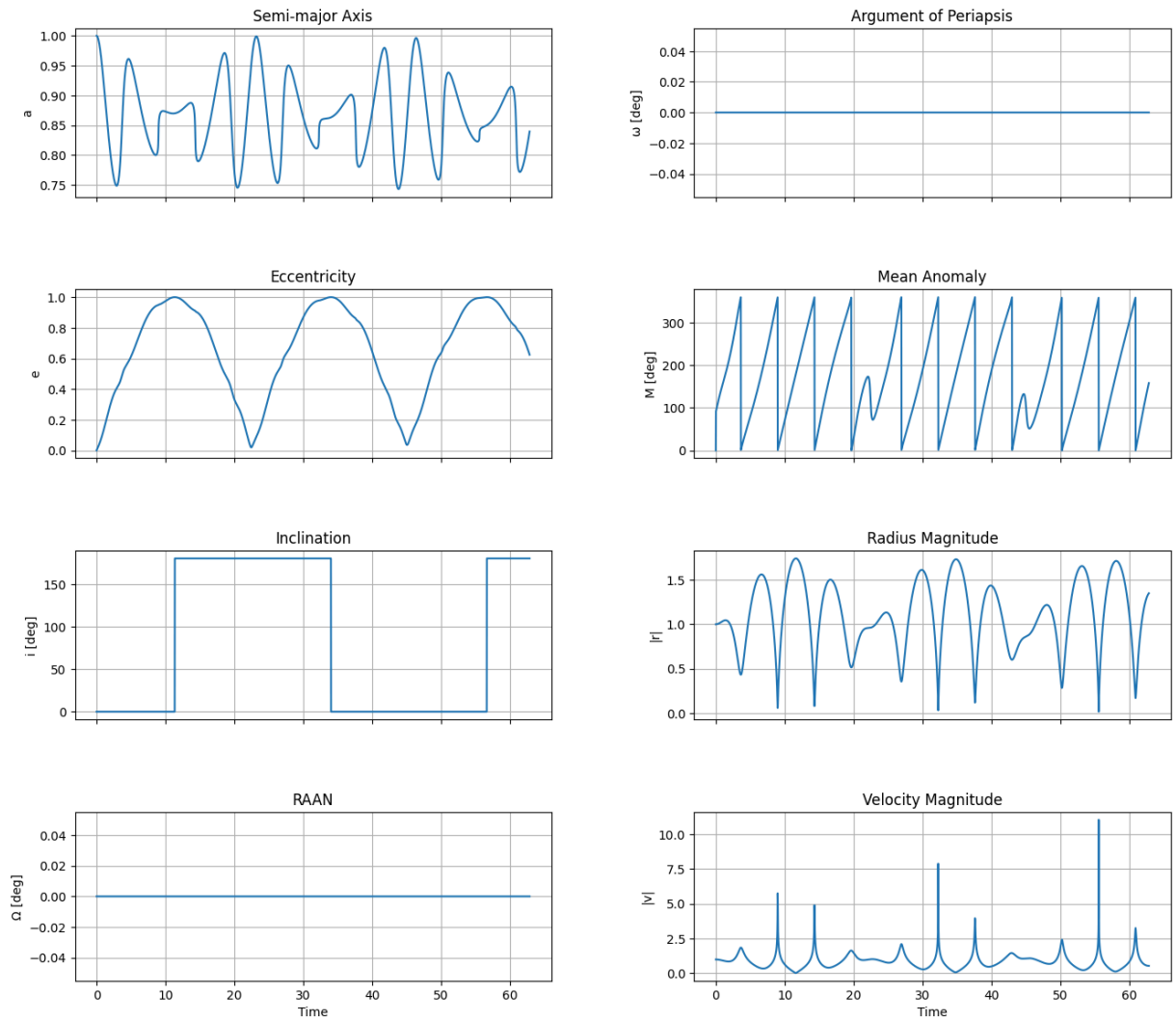
Running case:  $a_d = [0.01, 0, 0]$

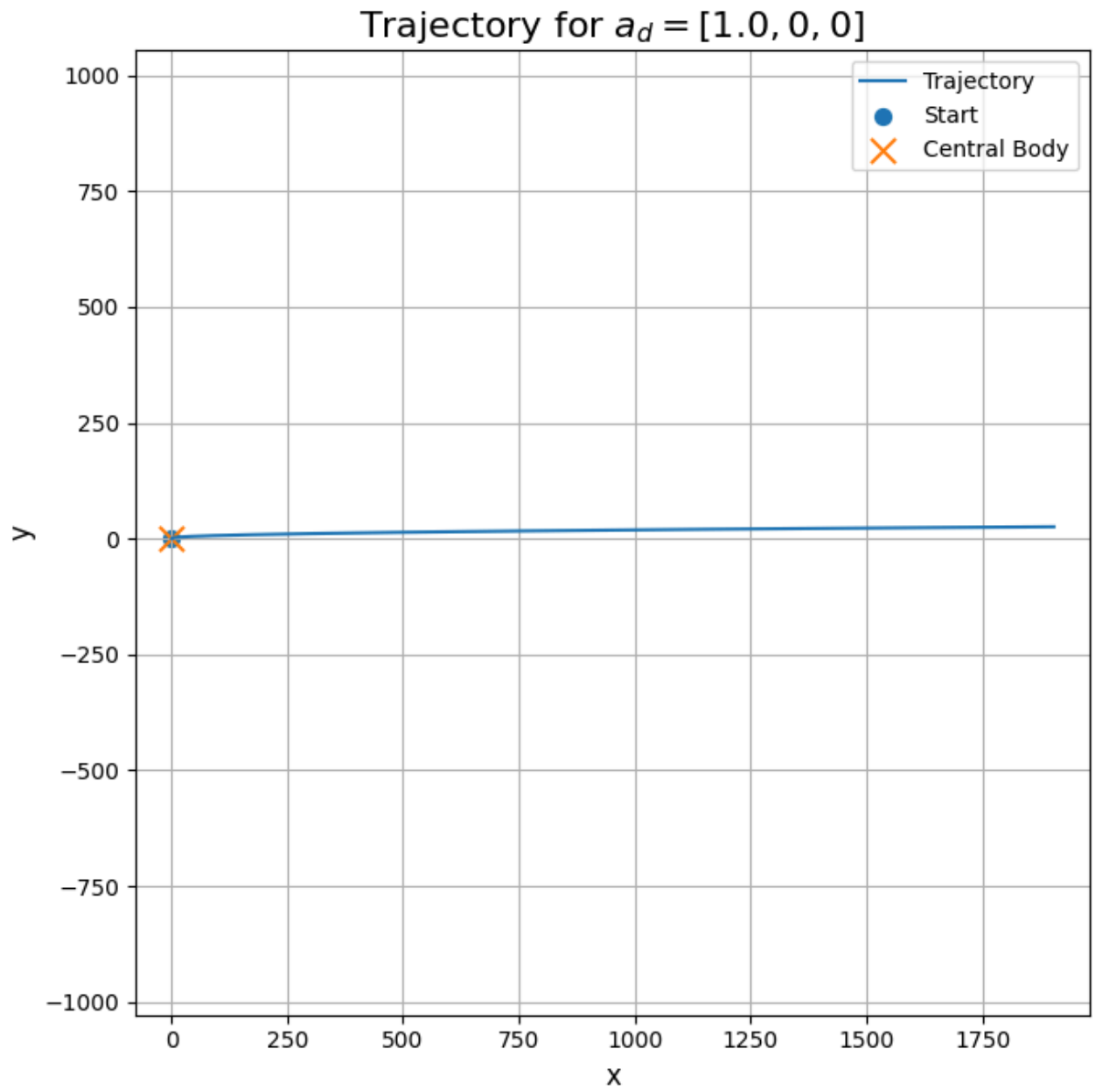




Orbital Elements vs Time for  $a_d = [0.01, 0, 0]$ Running case:  $a_d = [0.1, 0, 0]$



Orbital Elements vs Time for  $a_d = [0.1, 0, 0]$ Running case:  $a_d = [1.0, 0, 0]$



Orbital Elements vs Time for  $a_d = [1.0, 0, 0]$ 